

Review of “Evidence Information Gain for LLM Confidence Estimation: A Diffusion Decision Model-Inspired Approach”

Mock ICLR review (paper only, no code access)

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Part I: Review

Summary

The paper estimates the confidence of a reasoning LLM by treating its chain of thought as a sequence of evidence. The CoT is cut into K equal-length segments; for each segment the paper estimates how much the segment raises the model’s log-probability of its final answer, via either the loss of sampled continuations (white-box) or the agreement rate of 10 sampled continuations (black-box). Levels and first differences of these two signals give a $4K$ -dimensional vector, and a logistic regression maps it to a correctness probability. On Qwen3-8B and DeepSeek-R1-Distill-Qwen-7B, with a leave-one-domain-out protocol, the method reports AUROC ≈ 0.83 – 0.84 against 0.69 – 0.71 for 10-sample self-consistency and 0.78 for a budget-matched self-consistency.

Strengths

- Simple, clearly motivated idea; the pipeline can be reproduced from the description at the level of its main components.
- Cross-domain hold-out is a harder and more meaningful protocol than random splits.
- Large gains over SC-10, consistent in every held-out group in the appendix tables, on both models.
- A budget-matched self-consistency baseline is included, and the black-box variant (EIG-SC) closely tracks the full model, a useful practical result.
- The K ablation is reported honestly: performance is flat in K .

Weaknesses

1. Cost is never stated, and the budget comparison is thin. By the paper’s own description the method needs 10 continuations per segment, so 50–250 generations per question. No table gives token cost. SC-Budget exists for Qwen3-8B only; DeepSeek’s entry is “—” with no explanation. In Fig. 4(a), SC-Budget sits at ≈ 0.78 for every K while CoT-EIG-Sum drops to ≈ 0.77 for $K \geq 10$, so the sentence “comparable to CoT-EIG-Sum” understates it: at matched budget only the Mean variant wins, by roughly 0.03 – 0.06 AUROC. The abstract’s “consistently outperforms existing methods” should be read against that number.

2. What are the “50 independent runs”? Table 1 and §4.3 say experiments are “repeated 50 times”. Repeated how: different training seeds, different rollout samples, different splits? The logistic regression and the hold-out set are both deterministic as described, so it is not clear what varies. If it is a bootstrap, say so; the $\pm$ then means something different.

3. Model diversity. Both models are Qwen-derived (the DeepSeek model is a distillation onto Qwen2.5-7B) and both are 7–8B. “Multiple LLMs” is two closely related ones. No larger or non-Qwen model; no closed model despite the black-box claim.

4. No trajectory-level baseline. Related work discusses Mao et al. (2025, 2026), Grünefeld et al. (2026) and Xiong et al. (2026) as the closest work; none is compared to. $P(\text{True})$ and verbalized confidence are also absent. Sum-Prob is listed in §4.2 but does not appear in Table 1.

5. ECE for the baselines is not meaningful. Non-probabilistic scores are min–max normalized (§4.1), which is not a calibration and uses the test set’s own extremes. The ECE column therefore compares a fitted probability (ours) with an unfitted rescaling (theirs). A fair comparison would Platt-scale the baselines on the same training benchmarks the logistic regression uses.

6. Method details are under-specified or inconsistent.

- Eq. 2 is stated for Y and cites A.1, but A.1 proves the identity for $Y_{>i}$. The replacement of Y by $Y_{>i}$ is said to “preserve the interpretation” without argument.
- “ ≥ 0 ” in Eqs. 2–3 holds for the mutual information, not for the sampled estimator, which can be negative; the features presumably are.
- Eq. 4 keeps only continuations that reach “the target answer Y ”. Which Y : the model’s own final answer or the gold answer? What happens when no continuation reaches it ($N_{\text{succ}} = 0$)? And A.2 derives the plain $1/N$ average, not the success-filtered one in Eq. 4, so the appendix does not prove the equation it is cited for.
- The self-consistency branch uses an agreement probability “as a single estimate of $\mathbb{E}[\log p(\cdot)]$ ”; a probability is not an estimate of an expected log-probability.
- e_{loss}^i and e_{sc}^i are never written as formulas; the reader has to infer that e_{loss}^i is Eq. 4. Δ^1 is undefined.
- Segments are equal-length token chunks (§3.3), but the abstract and introduction speak of “reasoning steps”. These are not the same thing.

7. The DDM is a metaphor, not a model. No drift rate, boundary or starting point is fitted; “drift rate” is simply another name for conditional mutual information. Section 3.1 could be one motivating paragraph. As written, the title promises more than the method delivers.

8. Feature-contribution analysis is weak evidence. The tree in Fig. 2 predicts the regression’s own output, not correctness, so it explains the estimator rather than the phenomenon. Table 2 reports the largest importance (0.64) for self-consistency at evidence 1, the earliest segment. That suggests much of the signal is “self-consistency after the first fifth of the trace”, which is in tension with the evidence-accumulation story. The paper claims SC features “appear in more internal split nodes” but Table 2 reports Gini importance, not split counts, and Fig. 2 (depth 4) shows only a few splits; whether Table 2 is computed on the same depth-4 tree is not stated.

9. Missing experimental details. Number of questions per benchmark and per held-out set; how correctness is judged (exact match? LLM judge? especially for TruthfulQA in generation mode); max generation length; regularization and feature scaling for the logistic regression; the temperature of the SC-10 rollouts; which token LastRep uses. Countdown is listed as one of eight benchmarks but never appears in any table or figure.

10. “Black-box” is stronger than what is shown. EIG-SC needs the reasoning trace and the ability to resume sampling from a truncated trace at $T = 1.0$. Commercial reasoning APIs expose neither. “Requires no logits or hidden states” is the accurate claim.

Presentation

- The main text runs past the page limit (the conclusion continues on page 11).
- The conclusion is §5.4 under “Ablation Study”. Appendix A has two subsections with no titles, and Table 2 is placed inside “Proofs”.
- Several citations are textual where parenthetical is needed (“cumulative log-probability Ulmer et al. (2024)”).
- “each evidence” / “evidences” throughout; “a LLM”; “estimation.Six”; the §5.3 sentence beginning “Although CoT-EIG already combines...” is ungrammatical.
- Fig. 2 is on page 4 but first discussed on page 8. Fig. 4’s caption says “baselines” though it includes the proposed method.
- Table 1 bolds 0.84 over 0.83 with ± 0.01 on both.

Questions for the authors

1. What exactly varies across the 50 runs?
2. How does CoT-EIG compare with self-consistency computed on the first $1/K$ of the trace only, given Table 2?
3. What is the total generation cost per question for CoT-EIG vs. SC-10 vs. SC-Budget, and why is SC-Budget missing for DeepSeek?
4. In Eq. 4, is Y the model’s own answer or the gold answer?

Rating

5: marginally below the acceptance threshold. Confidence: 4. The empirical gain over self-consistency is real and the black-box result is useful, but the cost accounting, the unclear “50 runs”, the weak baseline calibration, and the two-Qwen-model evaluation leave the central claim (“consistently outperforms existing methods”) under-supported. I would raise to 6 with clear cost and protocol reporting and an honest matched-budget statement; a third, non-Qwen model would move it further.

Part II: Suggested edits (paste-ready)

These address every text-level point above without new experiments. Numbers are taken from the logs and code and are exact. Ordered by priority.

1. Table 1 caption (tables/full_auroc_ece.tex)

Where: replace “Results are reported as mean $\pm$ standard deviation over 50 independent runs.”

Results are reported as mean $\pm$ standard deviation over 50 bootstrap resamples of the held-out evaluation set, averaged over the four held-out groups of Section~\ref{sec:cross_benchmark_generalization}.

2. §4.3, first paragraph

Where: replace “Each benchmark is used once as the target, ...reported across runs.”

This yields four held-out groups: the multiple-choice knowledge domain (MMLU and MMLU-Pro), the mathematics domain (GSM8K, MATH500 and AIME), GPQA, and TruthfulQA. Countdown is used for training only, since both models answer it almost perfectly and it does not contain enough incorrect answers to evaluate a confidence estimator. Reported numbers are the average over the four groups. Standard deviations are computed by resampling each held-out set with replacement 50 times and evaluating the fixed, already-trained estimator on each resample; they therefore reflect evaluation-set uncertainty rather than training variance.

3. §4.1 Models and Benchmarks

Where: append to the paragraph.

After generation, 575 and 532 questions are retained for Qwen3-8B and DeepSeek-R1-Distill-Qwen-7B, respectively (100--300 per benchmark). Per held-out group, the evaluation sets contain 551/538 (multiple-choice knowledge), 432/431 (mathematics), 185/157 (GPQA) and 298/299 (TruthfulQA) questions for the two models, with the estimator trained on the remaining 000--1,400 questions.

4. §4.2 Token-level baselines

Where: replace “and token probability (Mean-Prob, Sum-Prob) are considered.”

and the length-normalized sequence probability (Mean-Prob) are considered. Since lower loss and lower entropy indicate higher confidence, these scores are negated so that larger values correspond to higher confidence.

5. §4.2 Self-consistency baselines

Where: replace the whole paragraph.

\textbf{SC-10} uses the 10 rollouts sampled from the question alone (i.e., before any evidence is observed) and takes the fraction of rollouts that agree with the majority answer as the confidence. To provide a computationally matched comparison, \textbf{SC-Budget} increases the self-consistency sampling budget to \$25\$, \$50\$, \$75\$, \$100\$, and \$125\$ rollouts for $K \in \{5, 10, 15, 20, 25\}$, respectively, where K denotes the number of evidences in the chain-of-thought reasoning trace. These budgets approximately match the number of rollouts our method requires. SC-Budget was run for Qwen3-8B only; the corresponding DeepSeek-R1-Distill-Qwen-7B entries are therefore left empty.

6. §4.2, new paragraph after “Proposed method”

\paragraph{Computational cost.}

For each question, CoT-EIG requires one complete response and $10(K+1)$ sampled continuations: 10 from the question alone (shared with SC-10) and 10 from each of the K evidence prefixes. With $K=5$ this is 60 continuations, compared with 10 for SC-10 and 25 for SC-Budget. Since later prefixes are longer, the continuations are on average shorter than a full response. All continuations are generated once and reused by every variant and feature set, so the ablations in Section~\ref{sec:ablation_features} share the same rollouts.

7. §4.2, new paragraph after “Computational cost”

\paragraph{Implementation details.}

Responses are generated with a limit of 15{,}000 new tokens; continuations use the same limit. Final answers are extracted from the response and compared with the reference by exact match, using \texttt{math-verify} for the mathematical benchmarks. The self-consistency estimate for evidence i is the Laplace-smoothed fraction $(m_i+1)/11$ of the 10 continuations from C_i whose final answer equals the model’s own final answer Y , where m_i is the number of agreeing continuations; for $i=1$ the difference features Δ^1_{loss} and Δ^1_{sc} are set to zero. The confidence estimator is an ℓ_2 -regularized logistic regression (scikit-learn defaults, $C=1$) trained on the unscaled feature vector. \textbf{LastRep} uses the final-layer hidden states averaged over the generated tokens as its input. All experiments use vLLM on a single GPU.

8. §5.2, second paragraph

Where: replace the paragraph.

Figure~\ref{fig:evidence_ablation_models} shows that, on Qwen3-8B, \textbf{SC-Budget} reaches an AUROC of about 0.78\$ at every budget, which is above \textbf{SC-10} but 0.03\$--0.06\$ below \textbf{CoT-EIG-Mean} across all K , and comparable to or slightly above \textbf{CoT-EIG-Sum} for $K \geq 10$. Thus, at a matched number of rollouts, the gain of the proposed method comes from CoT-EIG-Mean, and it is moderate but consistent across K .

9. Black-box wording

Where: abstract, replace the last sentence.

Importantly, a variant of our method requires only sampled continuations, without access to token probabilities or hidden states, and achieves confidence estimation performance very close to that of the white-box variant.

Where: contribution bullet 1, replace “and can be applied to LLMs in a black-box setting, without requiring access to their internal representations or parameters,”

and has a variant that requires neither token probabilities nor internal representations, only the ability to sample continuations from a partial reasoning trace,

Where: §3.3, add after “...does not require access to the model’s loss or token-level probabilities.”

It does, however, require sampling continuations from a truncated reasoning trace, which we assume the deployment interface allows.

10. §4.4, second paragraph and Table 2 caption

Where: replace the second paragraph of §4.4.

As shown in Figure~\ref{fig:feature_importance_tree}, features from multiple evidences are used as splitting criteria, indicating that confidence estimation relies on information distributed across the reasoning trace. Table~\ref{tab:feature_importance} reports Gini importance for a fully grown tree fitted on the same data (maximum depth 20, minimum leaf size 1), whereas Figure~\ref{fig:feature_importance_tree} shows a separate tree limited to depth four and a minimum leaf size of 10 for readability. Self-consistency-based features account for roughly 90\% of the total importance, suggesting that they provide the stronger signal. The single most important feature is the self-consistency of the first evidence, i.e., agreement measured after the first $1/K$ of the trace; the remaining evidences add a further ≈ 0.3 of importance, indicating that the trajectory refines an early signal rather than replacing it.

Where: tables/feature_evidence_importance.tex, replace the caption.

\caption{Gini feature importance of a fully grown decision tree (maximum depth 20, minimum leaf size 1) fitted to predict the high/low-confidence label of CoT-EIG-Mean with $K=5$ on the combined data of all benchmarks for Qwen3-8B. Features with zero importance are omitted.}

11. Method section (§3.2, §3.3, Appendix A.2)

Where: §3.2, replace “we replace the final answer Y with $Y_{>i}$, the continuation of the reasoning chain from e_i through the final answer. This preserves the underlying interpretation while enabling efficient estimation”

we replace the final answer Y with $Y_{>i}$, the continuation of the reasoning chain after e_i up to and including the final answer. Since Y is a deterministic function of $Y_{>i}$, the data-processing inequality gives $I(Y_{>i}; C_i \mid C_{1:i-1}) \geq I(Y; C_i \mid C_{1:i-1})$, so the tractable quantity upper-bounds the original one while retaining its interpretation and enabling efficient estimation

Where: §3.2, replace “considering only trajectories that ultimately reach the target answer Y .”

considering only continuations whose final answer equals Y , the final answer of the model's own original response. When no continuation reaches Y ($N_{\text{succ}}=0$), the estimate is undefined and the corresponding feature is set to zero.

Where: §3.3, replace “use the resulting probability as a single estimate of $\mathbb{E}[\log p(Y_{>i} | C_i)]$.”

use the resulting agreement rate in place of the loss-based estimate of $\mathbb{E}[\log p(Y_{>i} | C_i)]$ as the evidence-level signal.

Where: §3.3, add after Eq. 6.

with $\Delta_{\text{loss}} = \Delta_{\text{sc}} = 0$.

Where: §3.2: both “a LLM” $\rightarrow$ “an LLM”.

Where: Appendix A.2, replace the opening “By the chain rule:”

Eq.~\ref{eq:monte_carlo_estimator} restricts the average to continuations whose final answer equals Y ; the derivation below is for the unrestricted average, and the restricted form follows by replacing N with N_{succ} and the sum with the indicator-weighted sum. By the chain rule:

12. §3.1, shorten to one paragraph (frees about half a page)

Where: replace the three paragraphs of §3.1.

The Diffusion Decision Model (DDM)~\citep{ratcliff1978theory,ratcliff2008diffusion} describes decision making as sequential evidence accumulation under noise, in which a decision is reached when accumulated evidence crosses a boundary. Its central parameter, the \emph{drift rate}, captures the rate and direction of accumulation and reflects the quality of the available evidence~\citep{ratcliff2002estimating,ratcliff2016diffusion}: strong evidence drives fast, consistent accumulation, whereas weak or ambiguous evidence produces slow and erratic accumulation. We borrow this perspective, not the model itself: we treat the segments of a reasoning trace as evidence and quantify each segment's contribution to the final decision, without fitting drift, boundary, or starting-point parameters.

13. Structure and wording

- \subsection{Conclusion} $\rightarrow$ \section{Conclusion}.
- Appendix: first \subsection{} $\rightarrow$ \subsection{Derivation of Eq.~\ref{eq:drift_rate_requation_2}} second $\rightarrow$ \subsection{Derivation of Eq.~\ref{eq:monte_carlo_estimator}}.
- Move \input{tables/feature_evidence_importance} under a new \section{Feature Importance} after the proofs.
- \cite{ } $\rightarrow$ \citep{ } in all 7 places (§2, §3.3 wei2022chain, §4.1).
- §5.3: “estimation.Six” $\rightarrow$ “estimation. Six”.
- §5.3, replace “Although **CoT-EIG** already combines ...as shown in Figure 5.” with:

\textbf{CoT-EIG}, which combines the loss- and self-consistency (SC)-based features, achieves the strongest overall performance in both AUROC and ECE, while the EIG-SC variant performs very close to it, as shown in Figure-\ref{fig:feature_ablation}.

- Figure 4 caption: “Performance of different confidence estimation baselines” → “Performance of the proposed method and the baselines”.
- Abstract and contribution bullet 3: “eight diverse benchmarks and multiple LLMs” → “eight benchmarks and two reasoning models”.

14. Limitations sentence

Where: conclusion, insert before the “More broadly” paragraph.

Our evaluation covers two 7--8B reasoning models that share a Qwen lineage and a fixed rollout budget of 10 continuations per evidence; extending the comparison to larger and unrelated model families, and to recent trajectory-level estimators~\citep{mao-etal-2026-confidence,grunefeld2026tracing}, is left for future work.